

Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



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What did we see?



When does it work?

| Direct (practical) extensions

- ◇ Algorithm design
- ◇ Lyapunov
- ◇

| Interpolation

- ◇ Many (similar classes)
- ◇ Linear operators

| A few success stories

- ◇ Algorithm analysis

- ◇

| A few success stories

- ◇ Algorithm design

- ◇

| A few instructive examples

Design first-order methods via PEPs:

- ◇ Drori, Teboulle (2014). "Performance of first-order methods for smooth convex minimization: a novel approach."
- ◇ Kim, Fessler (2016). "Optimized methods for smooth convex optimization."
- ◇ Van Scoy, Freeman, Lynch (2017). "The fastest known globally convergent first-order method for minimizing strongly convex functions."
- ◇ Kim, Fessler (2021). "Optimizing the efficiency of first-order methods for decreasing the gradient of smooth convex functions."
- ◇ Altschuler, Parrilo (2023). "Acceleration by Stepsize Hedging I: Multi-Step Descent and the Silver Stepsize Schedule."
- ◇ Jang, Das Gupta, Ryu (2025). "Computer-assisted design of accelerated composite optimization methods: OptISTA."
- ◇ Bok, Altschuler (2025). "Optimized methods for composite optimization: a reduction perspective."

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- ◇ Bok, Altschuler (2025). "Optimized methods for composite optimization: a reduction perspective."

... including "brutal" examples:

- ◇ Grimmer (2024). "Provably faster gradient descent via long steps."
- ◇ Gupta, Van Parijs, Ryu (2024). "Branch-and-Bound Performance Estimation Programming: A Unified Methodology for Constructing Optimal Methods."
- ◇ Kamri, Hendrickx, Glineur (2025). "Numerical Design of Optimized First-Order Algorithms."

| A few (more complicated) directions

- ◇ nonconvex PEPs (e.g., adaptive stepsizes)
- ◇ nonconvex PEPs (e.g., second-order, low-dim)
- ◇ higher-order PEPs
- ◇

| Community work

- ◇ Maintain & improve packages
 - PESTO, PEPit, PEPit.jl
 - AutoLyap, AutoLyap.jl
- ◇
- ◇